國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

Lab 01

姓名：林莉婕
學號：112AC1015

Lab01-Ex1. (題目編號)
印出**Hello World!**，將過程及結果按照標準格式 呈現後轉成**pdf**，並上傳**Google Classroom**
Code

```
#include <iostream>

int main()
{
   std::cout<<"Hello World"; //印出hello world

   return 0;
}
```

Lab01-Q#. (題目編號)		
Briefly describe the problem. (題目簡述)		
Code		
Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	Paste your code here. Please use Pygments to highlight your code. (請使用Pygments傳換程式碼格式後再貼上)	
result		
(a)	(b)	(c)
	Show your result in tables or screenshots. (請使用表格或是截圖呈現結果。)	
Discussion		
Discuss and conclude your results, or answer questions. (實驗討論、結論或回答問題。)		

[EXAMPLE] Q1. Factorial Function

The factorial function n! has value 1 when n<=1 and value n*(n-1)! when n>1.

Please write both a recursive and an iterative C function. Indicate the input, output, and give proper comments. Test your codes and log the results.

Code

<iterative version>

```c
#include<stdio.h>
#include <stdlib.h>
int main()
{
int n,answer;
char i;
printf("Factorial function n!,n=");
scanf("%d",&n); //輸入要查詢的 n
if(n<=1)
{
printf("Answer=1"); //小於 1 時答案為 1
}
else //大於 1 時
{
answer=n; //計算 (n>12 時超出 int 儲存上限)
for(i=(n-1);i>1;i--)
{
answer=answer*i;
}
printf("Answer=%d\n",answer);
}
system("pause");
return 0;
}
```

<recursive version>

…

result

```
Factorial function n!,n=7
Answer=5040
```

Discussion